



Clean Water. Smarter Systems.

SOUTH TEXAS WATER RESPONSE | NEAR-TERM SUPPLY & SYSTEM OPTIMIZATION

South Texas is facing a rapidly tightening water supply driven by prolonged drought, accelerating industrial growth in Corpus Christi, and increasing reliance on the Evangeline Aquifer. Current response efforts focus on seawater desalination (6–10 year timeline) and expanded groundwater extraction (2–4 years) — **these solutions are necessary but do not address immediate supply needs.**

South Texas has 60–75 MMGPD of recoverable water currently being underutilized — a significant near-term supply opportunity.

Key Observation

Corpus Christi processes ~120 MMGPD. Of this:

- ~40–50 MMGPD discharged as wastewater effluent
- ~20–25 MMGPD is process water suitable for recovery

Total recoverable: **60–75 MMGPD**

Community Impact

Large-scale groundwater withdrawals are expected to:

- Lower water tables and increase salinity
- Degrade private well systems in communities like Goliad
- Create a regional equity issue between urban demand and rural resources

Purionex — Modular. Deployable in Months. Lower Cost than Conventional Desalination.

IMMEDIATE (0–6 Months)	NEAR-TERM (Municipal Augmentation)	RURAL (Community Support)
Recycling & Recovery • Deploy modular systems at industrial and municipal sites • Recover wastewater and process water • Reduce demand on municipal supply	Municipal Augmentation • Treat groundwater to meet or exceed existing inlet water quality • Integrate into current treatment plants • Lower operating cost	Rural Community Support • Enable small communities to treat local groundwater • Protect well systems and maintain local control • Independently deployable

KEY ADVANTAGES: Speed — deployment in months, not years (16–20 weeks for full-scale units) · Cost — lower capital and operating cost vs. conventional desalination · Scalability — modular 1 MMGPD units expandable as needed · Integration — works with existing infrastructure

State Coordination Opportunity

To accelerate deployment, we recommend:

- **Regulatory Alignment:** Coordination with TCEQ to establish expedited pathways for pilot projects and water augmentation approvals
- **Funding Support:** Utilize Texas Water Development Board programs and USDA Rural Development
- **Regional Coordination:** Align municipalities, industry, and rural communities on recycling and efficiency

Why This Matters Now

Long-term projects remain essential — but they require years and significant capital. Purionex provides a near-term bridge:

- Delivers meaningful new water supply in months
- Reduces dependence on groundwater withdrawals
- Lowers total system cost
- Protects rural communities

What We Are Asking

State of Texas coordination to:

Support TCEQ expedited pathway for pilot projects • Align Texas Water Development Board funding • Coordinate municipal, industrial, and rural stakeholders • Prioritize near-term recycling solutions alongside long-term supply development

Texas leads on water. Solutions are ready. Now we act.

CONTACTS

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